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Reconfiguring Inherited Service Infrastructures for Green Public Service Transition

Municipally owned arrangements in Helsinki, Aalborg and Skåne

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Green transition is decided one investment at a time

The setting

- Municipalities seek lower emissions and more resource recovery, yet homes must stay heated and waste treated during the change.
- Existing networks offer a route to new production technologies, but replacing plants or adding recovery requires capital and dependable income.
- A climate target gives a reason to invest; it leaves technology, expenditure and responsibility to owners and companies.

Why municipally owned enterprises

- The owner can instruct the company and support borrowing, while depending on distributions and facing demands for affordable services.
- Managers weigh reliability, returns and debt service. Investment decisions make this relationship observable.

Research question

How do political objectives and business appraisal interact in municipal green transition investment, and how do institutions and ownership practices shape that interaction?

What we look at

How public owners respond when a politically supported investment still faces an adverse appraisal or an unsettled revenue basis: **adapting a production pathway, allocating costs through a supply agreement, qualifying support through financing and purchasing conditions.**

Analytical framework: how investability forms

Institutional conditions rules and established practices: authority · cost recovery · purchasing conditions

Political objectives
publicly stated reasons for intervention

tension or alignment

Business appraisal
costs, revenues, financing, operating requirements

municipal councils and company boards: appraisal and authorisation

Owner capital and guarantees
financing support

Prospective owner purchases
conditional revenue

Agreements between companies
cost allocation

Approved financing + **Credible cost recovery**

= Investability *the definition used in this study*

Investability

Approved financing combined with credible cost recovery. It asks who accepts the financial commitments and how investment and provision costs are expected to be met, rather than scoring completed business cases.

Revisability (secondary)

Changes in technology, scale, timing or activities: can the two conditions be maintained through the change?

Source: Authors' own elaboration (Figure 1 of the paper). Both terms are analytical tools, following Bruton et al. (2015) rather than assigning political and economic rationality to separate camps.

Where the paper sits

Municipal climate governance

Service provision as a means of implementing climate policy (Kern & Alber 2009). Mayors also back renewables for employment, earnings and local development (Busch & McCormick 2014).

In this paper: *We therefore record the purposes invoked for each investment.*

Public service ecosystems and public value

Value creation mediated among actors, institutions and competing values (Osborne et al. 2022; Trischler et al. 2023); conflict over public value foregrounded (Skålén et al. 2024).

In this paper: *We examine that mediation at the moment of investment, when purposes meet assets, expenditure and obligations.*

Municipally owned enterprises, multiple principals

Hybrid organisations combining public control and corporate form (Bruton et al. 2015). Joint delivery raises collective action problems (Voorn et al. 2019; Voorn & van Genugten 2022).

In this paper: *We ask whose agreement is required and how a collective position becomes an owner's commitment.*

Gap addressed: investment decisions are where an owner endorses a purpose, provides capital or considers purchasing an output, each with different consequences for the company's business case. Most studies stop at objectives; we follow expenditure, income and obligations.

Research design: three arrangements, fifteen documentary records

Helsinki · Helen Oy

H

District heat and power

City-owned company incorporated 2015 (Act 626/2013). Single owner; council and board decisions on production pathway and capital.

Aalborg · municipal heat group

A

District heat

Aalborg Forsyning A/S group: Aalborg Varme A/S buys heat from Nordjyllandsværket A/S. Council guarantees; intra-group agreements.

Skåne · Sysav

S

Waste treatment and resource recovery

Owned jointly by 14 municipalities since the 1970s. Malmö (45.98%) as the largest owner; subsidiary Sysav Industri AB.

Case logic

- Exploratory, purposive comparison (Seawright & Gerring 2008); not national models, not an isolated ownership effect.
- Case = the municipal service arrangement; observation = a documented investment decision sequence.
- Started from three substantive expectations (Finnish earnings and green investment, Aalborg mayoral leadership, Sysav's long cooperation) and kept the records that qualify them.

Documents and procedure

- 15 records (H1–H5, A1–A6, S1–S4): council minutes, annual reports, agreements, regulator guidance, CJEU material, press; Finnish, Danish and Swedish read in the original.
- Document analysis after Bowen (2009): authorship, purpose, proximity to the decision.
- Per episode: purposes invoked, economic assessment, action taken, actor and authority. Financing evidence and cost-recovery evidence read separately; a blank is a limit of the record, not proof of inaction.

Helsinki: political objective survives an adverse appraisal

H

Helen Oy, decentralised heat pathway and coal closures (records H1–H5)

Jun 2014
Council incorporates Helen; capital plan targets ≥50% equity through the 2016–2021 investment peak

2 Dec 2015
Council selects decentralised production (EUR 360m) over new CHP (EUR 960m) and Hanasaari closure

1 Apr 2023
Hanasaari B closed, after replacement heat capacity was secured

Apr 2025
Salmisaari, the last coal plant, closed; national coal ban from May 2029

2025 accounts
Fuel purchases EUR 336m → 216m; allowance costs EUR 78m → 25m (Helen 2026)

What the records show

- Board judged every option unattractive on purely commercial grounds; if the city acted for climate or land use, it preferred the less burdensome staged pathway.
- Capitalisation and plant selection were separate decisions: capital prepared first, pathway chosen later under uncertainty.
- Cost recovery tied to municipal finances through limits on heat-price rises and lower expected distributions; Consumer Ombudsman intervened on pricing (KKV 2022).

Reading

A politically authorised **modification of the pathway** through which public purposes would be pursued, incorporating the board's conditional preference.

The 2025 profitability is a **separate observation**: the accounts do not isolate the return on the 2015 choice from price movements.

Aalborg: who pays is settled by contract inside the municipal group

A

Aalborg Forsyning group, boiler agreement, network guarantee and the Vest heat pump (records A1–A6)

2015–2016
Municipality acquires Nordjyllandsværket from Vattenfall; parent guarantee for decommissioning

May 2017
Council approves a fossil-free strategy

9 Nov 2023
Vest 20 MW wastewater heat pump abandoned, attributed to the excess-heat price cap

4 Dec 2023
Regulator places wastewater heat outside the cap; the cap itself abolished July 2025

26 Jun 2025
Council guarantee DKK 385m for Varme's network: 17.3% of Varme's year-end assets

2025 reports
150 MW electric boiler assembled; Varme pays fixed capacity and variable costs at self-cost; commissioning summer 2026

What the records show

- The boiler agreement places the payment obligation on Aalborg Varme, the purchaser, through a supply agreement with the producer company.
- The guarantee exposes the municipality to borrowing risk; final risk assessment and guarantee commission are not reported in the published minutes.
- Danish heat pricing allows necessary costs into tariffs (hvile-i-sig-selv, §§20–21), subject to regulatory scrutiny.

Reading

Two interventions at **different corporate levels**: a municipal guarantee for one project and an intra-group cost allocation for another.

Vest shows a **recovery problem** attributed to a rule as then understood; the later correction shows why a company's appraisal must be read against the rule at the time.

Sysav and Malmö: support that keeps its conditions attached

S

SkyZero carbon capture and the Siptex divestment (records S1–S4)

Jul 1973 — Consortium agreement proposal links shared finance to dependable waste flows: capital, guarantees, voting, delivery duties

Apr 2024 — Board conditions for SkyZero: positive NPV on contracted revenues, sustainable financing, acceptable group risk, owner involvement

Autumn 2024 — 13 of 14 owner municipalities endorse the direction

Dec 2025 — Sysav decides to divest Siptex textile sorting after weak demand for sorted outputs; sale reported July 2026

26 Mar 2026 — Malmö authorises development and a guarantee ceiling ≈ SEK 1.2bn (45.98% of SEK 2.6bn)

2027 — Final investment decision envisaged; purchase of carbon removals conditional on procurement compliance

What the records show

- Projected revenues combine carbon-removal credits, avoided allowance costs and anticipated EU Innovation Fund and Industriklivet support; profitability described as dependent on support.
- Malmö's prospective purchase is tied to procurement law: in-house award (Teckal, Directive 2014/24 Art. 12) and the January 2026 CJEU judgment C-692/23 on group-level activity.
- A later reference (C-672/26) asks whether third-party sales count as entrusted activity; it is a question, not a ruling on SkyZero.

Reading

Malmö appears twice: as **financier** (guarantee ceiling) and as **prospective customer** (conditional purchase). Ownership did not remove the procurement condition.

Siptex is a **change of activities** after a weak market, distinct from abandoning a proposed investment.

Owner actions and the components of investability, by project

Condensed from Table I of the paper

Project (record)	Owner action	Financing basis	Cost-recovery basis	Stage / open condition
Helsinki: decentralised heat and coal closures (H1–H5)	Incorporation with ≥50% equity target (2014); council chose decentralised option (2 Dec 2015)	Prepared equity capacity; EUR 360m vs EUR 960m CHP	Board: options commercially unattractive; limited tariff pass-through. 2025: fuel and allowance costs down	Pathway authorised 2015; plants closed 2023 and 2025. Price vs production effects not separated
Aalborg: network and associated investments (A4)	Council guarantee DKK 385m (26 Jun 2025) with commission and risk assessment	Municipal guarantee for Varme borrowing; 17.3% of Varme assets	Statutory necessary-cost tariff route (§20); project costs' treatment not specified	Approved. Final risk assessment and commission not in cited minutes
Aalborg: 150 MW electric boiler (A3)	No direct owner action documented; intra-group agreement	Not documented	Varme pays fixed capacity and variable costs at self-cost	Cold testing done; commissioning summer 2026. Duration and continuity obligations unspecified
Aalborg: Vest 20 MW heat pump (A5–A6)	Not documented	Not documented	Abandonment attributed to excess-heat cap; regulator corrected 4 Dec 2023; cap abolished Jul 2025	Abandonment announced 9 Nov 2023. Later implementation not established
Sysav: SkyZero carbon capture (S2, S4)	13 of 14 owners endorsed direction; Malmö conditional authorisation and guarantee ceiling ≈ SEK 1.2bn; prospective purchases	Internal funds, anticipated Innovation Fund / Industriklivet, borrowing with owner support; conditional	Positive NPV on contracted revenues required; Malmö purchase conditional on procurement compliance and final approval	Development; FID envisaged 2027. Entrusted-activity treatment pending (C-672/26); grants unconfirmed
Sysav: Siptex textile sorting (S3)	Not documented	Not documented	Weak demand for sorted outputs	Divestment Dec 2025; sale reported Jul 2026. Post-sale financing not documented

Finding 1: owner support coexisted with adverse appraisals and open recovery conditions

Financial support was given, yet recovery stayed under scrutiny in every arrangement. The three records show the same gap in three forms.

Helsinki

H

Owner support

Capital prepared

Still outstanding / adverse

Board: all alternatives commercially unattractive

Council selected the board's conditional preference: a less burdensome, staged pathway. Political authorisation adjusted the means, not the purpose.

Aalborg

A

Owner support

Guarantee DKK 385m for network

Still outstanding / adverse

Boiler costs assigned to the purchaser by agreement

The group company accepted the producer's costs through a supply agreement. Support and payment obligations sit at different corporate levels.

Sysav / Malmö

S

Owner support

Guarantee ceiling ≈ SEK 1.2bn

Still outstanding / adverse

Contracted revenues and final approval still required

Malmö qualified its support: borrowing ceiling now, purchase subject to procurement compliance, decision in 2027.

Owner support advances a proposal; it does not by itself settle the business case.

Finding 2: institutions entered through appraisals and purchasing conditions

Tariff and price rules

- Helen: limits on heat-price increases in the 2015 appraisal; Consumer Ombudsman required changes to price-rise procedures (KKV 2022).
- Denmark: necessary costs may enter tariffs (§§20–21); Vest abandoned on the excess-heat cap as then understood, corrected a month later.

Appraised tariff capacity

Supply agreements between companies

- Aalborg Varme covers Nordjyllandsværket's fixed capacity and variable operating costs at self-cost for the 150 MW boiler.
- Duration and continuity obligations are not specified in the cited reports; the agreement identifies the payer, not the full recovery.

Contractual cost allocation

Procurement rules

- Malmö's purchase of carbon removals conditional on procurement compliance, final investment approval and legally effective decisions.
- In-house award (Teckal) requires >80% entrusted activity; Malmö linked its condition to CJEU C-692/23 (Jan 2026).

Conditional purchasing

Caveat: the presence of a rule does not establish that an investment's full costs will be recovered. The evidence of institutional influence lies in the specific assessment or condition, read against the rule as understood at the time.

Contribution and implications

To the multiple-principals literature

Agreement on direction can coexist with different positions on the obligations needed to act (refining Voorn et al. 2019; Voorn & van Genugten 2022). Agreement among public actors must be examined at the level of expenditure and income, not only objectives. Even within one municipal group, common ownership leaves the payment obligation to be specified by contract.

To public service ecosystem research

Mediation of public values here takes a material and financial form: which production option a council accepts, which company pays for heat, on what terms a municipality may buy carbon removals. Agreement on an environmental purpose coexists with disagreement about acceptable cost. Service continuity (replacement heat before closure) is a condition of the transition pathway.

On investability

Its value lies less in classifying projects as investable or not than in separating the commitments that make investment possible. Owner support did not settle recovery in Helsinki or SkyZero; Aalborg's agreement shows a purchaser taking responsibility for production costs. Incomplete records do not license the inference that municipal ownership relaxes financing constraints.

Managerial implication: make the scope of a commitment explicit

- the alternatives considered
- the party accepting each cost
- the revenue conditions
- the decisions still outstanding

This allows scrutiny of exposure while a common purpose stays politically supported. Where rules affect recovery or purchasing, their interpretation belongs in the appraisal.

Limitations and next steps

Limits of the evidence

- Documentary accounts capture formal decisions and reported explanations more readily than informal negotiation.
- Evidence is uneven across projects and stages; several financing or recovery cells remain blank.
- National effects are not isolated; the comparison identifies institutional channels in specific decisions, not country outcomes.
- Reported operating costs (Helen 2025) need a different assessment from the return attributable to an earlier choice.
- Revisability claims (Helsinki staging, Siptex sale) would need revised costs, revenues and accepted exposure from the same project.

Next steps

- Follow the same projects through implementation: boiler commissioning (summer 2026), SkyZero final decision (2027), post-closure accounts at Helen.
- Add the outstanding documents: Aalborg's final risk assessment and guarantee commission; the Varme–Nordjyllandsværket agreement's duration and continuity terms.
- Trace whether revisions preserved approved financing and credible recovery, and how accepted obligations changed during operation.
- Complement records with interviews on informal negotiation between councils, boards and group management.
- Test the framework on further arrangements (e.g. Finnish and German multi-utility groups) before any generalisation.

Central finding

Owner support can advance a politically endorsed proposal while commercial appraisal or revenue conditions remain adverse or unsettled.

Public ownership matters through the authority and organisational relationships used to negotiate these obligations. Its contribution to green transition lies in coordinating acceptable expenditure, financing and recovery, with choices about staging, contracts and conditional participation. Financing intervention alone cannot be treated as a settled business case.

Thank you

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Full paper, Table I and Appendix A (15 records) available on request.

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